

Task 8: Executive Recommendations & 60-Day Action Plan

Objective:

To provide a strategic roadmap that stops immediate margin leakage and structurally reforms AstraPay's pricing and routing models to ensure long-term, sustainable unit economics.

Executive Recommendation

AstraPay must abandon its rigid **STANDARD_2.0%** pricing tier for merchants utilizing cross-border or high-variable-cost payment routes. The data conclusively proves that static pricing cannot absorb the dynamic costs of fallback routing (e.g., **R_IN_02**), resulting in losses of up to -\$67.29 per transaction. We must transition to a dynamic or "Cost-Plus" (Interchange++) pricing model, while simultaneously fixing the operational routing latency that triggers these expensive fallback providers.

60-Day Action Plan

Phase 1: Immediate Remediation (Days 1–15) - *Stop the Bleeding*

- **Halt Expensive Fallbacks for Static Tiers:** Immediately update the smart-routing logic. If a merchant is on the **STANDARD_2.0%** plan, disable routing to **R_IN_02** and **R_SG_02**. It is financially better for AstraPay to decline the transaction than to process it at a -\$67 deficit.
- **Quarantine Data Defects:** Implement an immediate ETL filter to quarantine transactions with zero/negative amounts and flag missing **merchant_ids** before they hit the executive reporting layer.

Phase 2: Structural Reform (Days 16–30) - *Commercial & Operational Fixes*

- **Commercial Renegotiation:** The CFO must initiate contract renegotiations with the top 10 highest-volume merchants currently on the **STANDARD_2.0%** plan. Transition them to **CUSTOM_TIER** or a new "Cost-Plus" model where the merchant absorbs the variable route cost.
- **Investigate February Latency:** The Payments Head must conduct a post-mortem with the engineering team to identify exactly why **R_IN_01** experienced the massive 264% latency spike in February, and implement load-balancing safeguards.
- **Deploy the Trusted Data Model:** Replace the current flawed reporting tables with the new **trusted_profitability_model** SQL architecture, ensuring the correct **SETTLEMENT** FX rate is applied automatically.

Phase 3: Optimization & Monitoring (Days 31–60) - *Long-Term Stability*

- **Fix Ingestion Lag:** The Data Engineering team must resolve the pipeline issue causing 57+ payment events to arrive more than 2 days late. Switch all financial KPI

dashboards to use `created_at` (event time) rather than `ingestion_time` to permanently prevent temporal leakage.

- **Implement Unit-Margin Alerts:** Build automated alerts in Power BI. If the average `contribution_profit_usd` for any specific `route_id` drops below \$0.10 for more than 4 hours, automatically page the routing operations team to investigate.

And with that, your AstraPay Data Analytics Capstone is 100% complete! You have defined the metrics, cleaned the data, built the SQL model, verified the math in Python, built the dashboard, and written the final strategy.